

Conway's Law states:

"Organisations that design systems are constrained to produce designs that are copies of the communication structure of these organisations." This reflects the paradigm shift necessary for innovative solutions like blockchain to flourish.

In a federated blockchain or consortium blockchain like Hyperledger, where the participants are pre-selected and accepted, security and scalability can be managed in a controlled way. However, in a public or permissionless blockchain, anyone can participate, which makes it challenging to predict the scalability necessary for the blockchain system to function optimally.

Blockchain: The trilemma dilemma

The popular triple constraint theory has been elaborated and applied to blockchain by Vitalik Buterin as the 'blockchain trilemma'. It addresses the challenges technologists face in creating blockchain platform architecture that's decentralised, scalable, and secure.

A *decentralised* blockchain system empowers permissionless ownership, where anyone can use and build on the platform. Decisions with respect to transactions are made by way of consensus by a group of nodes. A consensus verified transaction is permanent and cannot be altered. Bitcoin, Ethereum, and other such cryptocurrencies suffer from slower performance as transactions grow, as they need to be confirmed by a single entity by design. There is light at the

end of the tunnel, though, in newer ways of implementation; one prime example is Ethereum 2.0.

Scalable blockchain systems are



Figure 1: Centralised, decentralised and distributed blockchain systems

able to handle growth in transactions gracefully. This is the most important aspect of any system for enabling mass adoption. However, this aspect of design is still in the works with respect to finding true solutions that would challenge current central authority-based scalability systems.

Secure blockchain systems self-manage their defence from attacks, bugs, and unforeseen issues. It's hard for blockchains to adhere to transparency at the source code level and still achieve a great amount of security. This is another area of work in progress.

Irrespective of whether one believes in the blockchain trilemma or not, it is true that building a reliable blockchain system that provides decentralisation, scalability, and security is a difficult undertaking for enterprises across all industries. In addressing these

challenges, multiple types of blockchains have evolved over the years.

Types of blockchains

What started as permissionless public blockchain technology has evolved into multiple types of blockchain combinations — public/private and permissionless/permissioned. An enterprise can choose the type of blockchain that inherently addresses most of its primary concerns as part of the framework.

While the blockchain has advantages like improved trust, security and speed; lower costs, and smart contracts, it does suffer from a few disadvantages. It is extremely difficult to modify data in a blockchain; it requires large database storage, and the private key is irrevocable even in case of loss. The selection between permissionless and permissioned blockchain becomes critical for an enterprise.

Although there are four types of blockchains, they largely boil down to public or private blockchains. The way in which these are deployed for management creates two other types of blockchains — consortium and hybrid. Public blockchains are permissionless and are decentralised peer-to-peer networks, whereas private blockchains are permissioned and the ledger is controlled by a centralised authority. Hybrid blockchains are part public and part private, with some parts of the system permission-based and some parts being permissionless. Lastly, consortium blockchains are federated with a set number of organisations

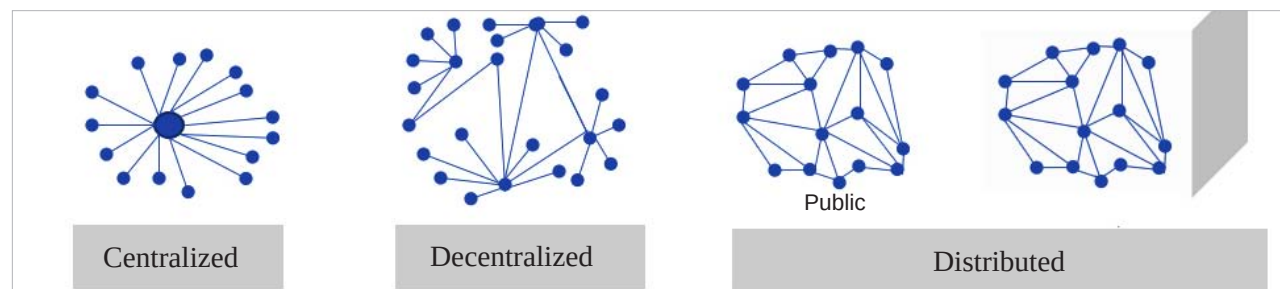


Figure 2: Blockchain types and characteristics